

QA Inspection Manual

MVTec AD Category-Based Industrial Inspection

For RAG Knowledge Base and AI Defect Detection Assistant

Manual ID: QA-MVTEC-AD-001

Version: 1.0

Category scope: bottle, cable, screw, transistor, tile, toothbrush, zipper, metal_nut

Purpose: This document provides a realistic QA inspection knowledge base for a student-built AI Defect Detection and Root-Cause Assistant. It is designed to be chunked, embedded, and retrieved by a RAG system when the AI needs inspection rules, acceptance criteria, and operator guidance.

Important disclaimer: This is a synthetic training/manual document. It is not an official MVTec document and not a certified manufacturing standard. In a real factory, replace this file with approved company QA manuals and SOPs.

Document Control

Field	Details
Manual ID	QA-MVTEC-AD-001
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Document type	Synthetic QA/SOP document created for student project, prototype, and RAG testing
Important note	This manual is not an official factory standard. It is a realistic simulated document for an AI inspection assistant. Human review is required for real industrial decisions.

1. Purpose and Scope

This QA Inspection Manual defines the inspection rules, acceptance criteria, anomaly indicators, and operator decision guidance for an AI-supported industrial inspection prototype using selected MVTec AD categories. The manual is written for use in a Retrieval-Augmented Generation knowledge base, where each section can be retrieved by product category, defect phrase, severity phrase, or operator question.

The manual supports the following selected categories: bottle, cable, screw, transistor, tile, toothbrush, zipper, and metal_nut. The system is intended to support image-based anomaly detection, visual explanation, evidence retrieval, and inspection report generation.

- Use this manual when the AI detects or suspects a visible anomaly in one of the selected product categories.
- Use this manual to explain whether a product should be accepted, reworked, rejected, or escalated to a human inspector.
- Use this manual to produce evidence-based report text for RAG outputs.
- Do not use this manual as the only source for real production safety or compliance decisions.

2. Source Basis and Dataset Context

The inspection assistant is designed around the MVTec AD dataset structure selected for the project. The official MVTec AD download page provides separate category downloads and describes the dataset folders as train, test, and ground_truth. In this project, the train folder contains normal/defect-free reference images, the test folder contains normal and anomalous samples, and the ground_truth folder contains pixel-level anomaly masks where available.

Item	Description
Dataset family	MVTec AD - industrial anomaly detection benchmark
Selected categories	bottle, cable, screw, transistor, tile, toothbrush, zipper, metal_nut
Training principle	Train on normal/defect-free examples first; test on normal and anomalous examples.
Project use	Visual anomaly detection, defect explanation, QA decision support, and RAG-based evidence retrieval.
Official source note	The dataset itself is provided by MVTec under a non-commercial Creative Commons license. Check the official license before sharing, publishing, or commercial use.

3. Key Terms and Definitions

Term	Meaning
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For RAG prototype use only - not an official industrial standard

Normal item	A product image that shows the expected accepted appearance with no visible anomaly.
Anomaly	Any visible deviation from the normal accepted appearance. In this manual, anomaly can include breakage, contamination, misalignment, missing part, surface damage, or abnormal texture.
Defect	An anomaly that affects quality, function, safety, appearance, or customer acceptance.
Acceptance criteria	The rule used to decide whether the item can be accepted, reworked, rejected, or escalated.
RAG evidence	A retrieved manual passage used to support the AI explanation and recommendation.
Confidence score	The model estimate of how likely the predicted result is correct. Confidence does not replace human judgement.
Ground truth mask	A pixel-level annotation used for evaluation of anomaly localization in the dataset.
Human review	Manual inspection by a trained person when the AI is uncertain, the image is unclear, or the defect is critical.

4. Roles and Responsibilities

Role	Responsibility
Operator / user	Upload clear product images, confirm product category, review AI result, and follow escalation rules.
Quality inspector	Verify uncertain, critical, or disputed AI predictions and make the final decision.
AI inspection assistant	Detect anomalies, retrieve relevant QA guidance, explain evidence, and prepare reports.
System administrator	Maintain model files, RAG documents, vector database, audit logs, and API keys.
Project developer	Ensure the AI response is grounded in retrieved documents and does not invent unsupported root causes.

5. Image Input Requirements

The AI system should inspect only images that are clear enough for visual analysis. Poor-quality inputs can cause false positives or false negatives. The AI assistant must ask for a new image when the input is not usable.

Input requirement	Inspection rule
Focus	Product edges, surfaces, connectors, threads, and small structures must be visible. Reject blurry images.
Lighting	Lighting should be stable and not too dark, too bright, or covered by strong glare.
Framing	The complete item or relevant inspection region should be visible. Cut-off products require retake.
Background	Background should not hide the product border or create confusing shadows.
Resolution	Use the original dataset image or highest available quality. Avoid compressed screenshots.
Category confirmation	The user should identify the category when possible: bottle, cable, screw, transistor, tile, toothbrush, zipper, or metal_nut.

Image rejection rule: If the product cannot be clearly identified, if the defect region is too blurry, or if the item is heavily cropped, the AI assistant must request a clearer image before providing a final QA decision.

6. General QA Decision Levels

The following decision levels are used across all categories. The exact decision can be adjusted by category-specific rules later in this manual.

Decision	Meaning	Action
Accept	No visible anomaly or only harmless appearance variation within normal accepted range.	Allow item to pass. Record as normal if required.
Rework / clean	Minor visible issue that can be corrected without reducing product function.	Send for cleaning, adjustment, or rework. Reinspect after correction.
Reject	Clear defect that affects safety, function, sealing, assembly, electrical connection, or customer appearance.	Remove item from accepted batch. Mark as defective.
Escalate	AI confidence is low, RAG evidence is missing, or defect may be critical.	Send to human inspector or supervisor for final judgement.

7. General Anomaly Indicators Across All Categories

- Broken, cracked, chipped, torn, or missing product structure.
- Unexpected contamination, dirt, stain, oil mark, glue mark, discoloration, or foreign material.
- Misalignment, deformation, bending, rotation, flip, or displacement of a part.
- Surface scratches, dents, pits, roughness, abnormal texture, holes, or material loss.
- Missing functional components such as cable wire, transistor lead, zipper tooth, screw thread, or toothbrush bristle region.
- Any anomaly that may affect sealing, assembly, electrical function, mechanical fit, hygiene, or customer safety.

Root-cause language rule: The AI may state possible or likely causes, but must not state a guaranteed cause from the image alone. Use wording such as possible cause, likely inspection checkpoint, or recommended process check.

8. Category-Specific Inspection Procedures

Each category section is written to be self-contained for RAG retrieval. When a user asks a question about a category or detected anomaly, the retriever should return the relevant category section and the general QA decision levels.

Bottle: Category-Specific QA Inspection Procedure

Product category: Rigid container / bottle inspection. Focus areas include neck, rim, side wall, bottom edge, surface cleanliness, and structural integrity.

Normal accepted condition. An accepted bottle image should show a complete bottle shape with no visible breakage, no large deformation, no contamination inside or outside the inspection area, and no damaged neck or rim.

Bottle - Inspection checkpoints

- Check the bottle rim and neck for chips, cracks, broken edge, or uneven shape.
- Check the side wall for large breaks, dents, crushed areas, or missing material.
- Check the interior and visible surface for contamination, dirt, stains, or foreign particles.
- Check whether the bottle outline is continuous and not interrupted by structural damage.
- Check for abnormal color regions that do not match normal bottle material or lighting variation.

Bottle - Reject or anomaly indicators

- Reject if a crack, broken large region, broken small region, chipped rim, or damaged neck is visible.

- Reject if contamination is visible inside a region that must remain clean for packaging or sealing.
- Escalate if a suspected mark may be reflection but the confidence is low.
- Do not accept any bottle with damage that may reduce sealing quality, hygiene, or pressure resistance.

Bottle - Caution notes for the AI assistant

- Transparent or reflective bottle surfaces can create highlights that look like defects. Treat unclear reflections as human review cases.
- Small lighting changes should not be reported as contamination unless a visible material mark or foreign object is present.
- A bottle anomaly near the rim or sealing surface is more serious than a cosmetic mark away from functional areas.

Bottle - Recommended operator action rules

Condition found	QA decision	Operator action
No visible anomaly	Accept	Allow item to pass inspection.
Surface contamination only, removable	Rework / clean	Clean item and reinspect.
Crack, chipped rim, broken side, damaged neck	Reject	Remove from batch and report structural defect.
Unclear reflection or low confidence	Escalate	Request human review or clearer image.

Bottle - RAG retrieval keywords

Use these phrases to retrieve this section: bottle crack, bottle broken, bottle contamination, bottle rim defect, bottle sealing issue, bottle reject criteria.

Bottle - Minimum evidence to include in report

- Product category and filename.
- Detected anomaly region: rim, neck, side wall, base, or internal surface.
- AI confidence score and whether human review is required.
- QA decision: accept, rework/clean, reject, or escalate.
- Short evidence sentence from this manual.

RAG answer instruction: When the retrieved section is Bottle, the AI assistant must explain the visible anomaly, state whether the item is acceptable or requires review, and give the recommended operator action. Do not state a guaranteed root cause unless process data confirms it.

Cable: Category-Specific QA Inspection Procedure

Product category: Flexible electrical cable / harness inspection. Focus areas include outer insulation, inner wire region, wire order, cable continuity, and exposed conductors.

Normal accepted condition. An accepted cable image should show complete cable insulation, correct cable placement, no missing wire, no cut insulation, no exposed conductor, and no abnormal bending that could affect function.

Cable - Inspection checkpoints

- Inspect outer insulation for cuts, holes, pokes, cracks, or exposed inner material.
- Inspect inner wire region for missing wires, swapped wires, bent wires, or incorrect cable arrangement.
- Check cable ends for clean termination and no broken strands.
- Check whether cable geometry follows the expected route without severe deformation.
- Check for contamination or foreign material attached to insulation.

Cable - Reject or anomaly indicators

- Reject if inner conductor or wire is exposed due to cut or missing insulation.
- Reject if a wire is missing, swapped, or arranged incorrectly for the expected cable pattern.

- Reject if the cable has severe poke damage, deep cut, or broken strand.
- Escalate if a bend may be normal flexibility but the model marks it as anomalous with low confidence.

Cable - Caution notes for the AI assistant

- Cable is naturally flexible, so slight shape variation is not automatically a defect.
- Cable-swap and missing-wire cases are functional defects and should be treated as high severity.
- Cut insulation can be safety-critical because it can expose conductor material.

Cable - Recommended operator action rules

Condition found	QA decision	Operator action
Normal flexible shape, no visible damage	Accept	Allow item to pass.
Small surface dirt on insulation	Rework / clean	Clean and reinspect.
Cut insulation, exposed conductor, missing wire, swapped wire	Reject	Remove from electrical assembly and notify QA.
Unclear bend or shadow around cable	Escalate	Request clearer image or manual continuity check.

Cable - RAG retrieval keywords

Use these phrases to retrieve this section: cable defect, cable cut insulation, missing wire, cable swap, bent wire, poke insulation, outer insulation damage.

Cable - Minimum evidence to include in report

- Cable section affected: outer insulation, inner insulation, wire order, conductor, or termination.
- Whether conductor exposure is visible.
- Whether a wire is missing or misplaced.
- Decision and recommended action.
- Human review status if cable geometry is ambiguous.

RAG answer instruction: When the retrieved section is Cable, the AI assistant must explain the visible anomaly, state whether the item is acceptable or requires review, and give the recommended operator action. Do not state a guaranteed root cause unless process data confirms it.

Screw: Category-Specific QA Inspection Procedure

Product category: Threaded fastener inspection. Focus areas include screw head, neck, thread top, thread side, straightness, and surface condition.

Normal accepted condition. An accepted screw image should show a complete head, straight neck, consistent thread pattern, no missing thread material, no deep scratches, and no deformation that prevents fitting.

Screw - Inspection checkpoints

- Inspect screw head for scratches, dents, missing material, or manipulated front shape.
- Inspect screw neck for scratch, deformation, or reduced diameter.
- Inspect thread side and thread top for broken, flattened, missing, or contaminated thread pattern.
- Check whether the screw axis appears straight and not severely bent.
- Check for surface discoloration, residue, or tool damage.

Screw - Reject or anomaly indicators

- Reject if thread damage may prevent correct tightening or mechanical fit.
- Reject if the screw head is damaged enough to prevent tool engagement.
- Reject if the screw neck is cracked, bent, or deeply scratched.
- Escalate if the anomaly is very small and only visible in the predicted mask but not visually clear.

Screw - Caution notes for the AI assistant

- Screw images may be grayscale in some workflows, so color-based judgment should not be required.
- Thread pattern contains many fine details; avoid overreacting to minor lighting shadows between threads.
- Any functional thread damage should be considered more serious than cosmetic surface variation.

Screw - Recommended operator action rules

Condition found	QA decision	Operator action
Consistent thread and no visible damage	Accept	Allow item to pass.
Minor surface mark away from thread/head	Rework / clean	Clean or mark for cosmetic review.
Damaged thread, broken head, bent shaft, deep scratch	Reject	Remove from assembly use.
Fine-detail uncertainty	Escalate	Send to human inspector with magnified image.

Screw - RAG retrieval keywords

Use these phrases to retrieve this section: screw defect, thread damage, scratch head, scratch neck, thread side defect, thread top defect, fastener reject.

Screw - Minimum evidence to include in report

- Screw region affected: head, neck, thread top, thread side, tip, or shaft.
- Whether tool engagement or thread fit may be affected.
- Confidence and anomaly mask/region if available.
- Decision and action.

RAG answer instruction: When the retrieved section is Screw, the AI assistant must explain the visible anomaly, state whether the item is acceptable or requires review, and give the recommended operator action. Do not state a guaranteed root cause unless process data confirms it.

Transistor: Category-Specific QA Inspection Procedure

Product category: Electronic component inspection. Focus areas include component case/body, leads/pins, alignment, placement, and physical integrity.

Normal accepted condition. An accepted transistor image should show an intact case, correct orientation, all leads present, no cut or bent leads beyond tolerance, and no misplaced body position.

Transistor - Inspection checkpoints

- Inspect the transistor case for cracks, chips, deformation, burn marks, or broken corners.
- Inspect all leads/pins for bending, cutting, missing material, or wrong spacing.
- Check whether the component body is placed and oriented correctly.
- Look for solder contamination, residue, foreign material, or damaged casing.
- Check whether the leads are usable for insertion, soldering, or electrical contact.

Transistor - Reject or anomaly indicators

- Reject if any lead is cut, missing, severely bent, or unable to make reliable electrical contact.
- Reject if the case is damaged, cracked, or displaced from the expected location.
- Reject if the component is misplaced or rotated in a way that could cause wrong assembly.
- Escalate if slight lead angle may be within tolerance but image evidence is unclear.

Transistor - Caution notes for the AI assistant

- Transistor defects are often functional, not only visual. Leads and placement are critical.
- A small cosmetic mark on the case may require review but is less severe than lead damage.
- Do not declare electrical failure from image alone; recommend electrical testing if physical defect is found.

Transistor - Recommended operator action rules

Condition found	QA decision	Operator action
Intact body and all leads normal	Accept	Allow item to pass visual inspection.
Minor removable residue	Rework / clean	Clean and reinspect.
Bent/cut/missing lead, damaged case, misplaced body	Reject	Remove from production and flag component quality.
Questionable lead angle	Escalate	Send to human review or electrical continuity test.

Transistor - RAG retrieval keywords

Use these phrases to retrieve this section: transistor defect, bent lead, cut lead, damaged case, misplaced transistor, electronic component QA.

Transistor - Minimum evidence to include in report

- Affected area: case, lead, pin spacing, placement, or orientation.
- Whether electrical contact could be affected.
- Recommended human or electrical test if needed.
- Decision and evidence phrase.

RAG answer instruction: When the retrieved section is Transistor, the AI assistant must explain the visible anomaly, state whether the item is acceptable or requires review, and give the recommended operator action. Do not state a guaranteed root cause unless process data confirms it.

Tile: Category-Specific QA Inspection Procedure

Product category: Surface texture / ceramic or patterned tile inspection. Focus areas include cracks, glue strips, stains, oil marks, gray strokes, rough patches, and visible texture disruptions.

Normal accepted condition. An accepted tile image should show a consistent surface pattern or texture, no visible crack, no unexpected glue line, no oil mark, no large stain, and no rough region outside normal texture variation.

Tile - Inspection checkpoints

- Inspect for cracks, broken lines, chipped texture, or fracture-like patterns.
- Check for glue strip, oil stain, gray stroke, discoloration, or foreign material on the surface.
- Check whether roughness appears as abnormal surface damage rather than normal tile texture.
- Look for local regions that strongly differ from surrounding texture.
- Check if the anomaly crosses a functional or visible surface area.

Tile - Reject or anomaly indicators

- Reject if crack, fracture, or surface break is visible.
- Reject if oil/glue contamination cannot be cleaned or affects customer-facing surface.
- Rework/clean if contamination is clearly removable and no material damage is visible.
- Escalate if model marks normal texture variation as anomaly with low confidence.

Tile - Caution notes for the AI assistant

- Texture images can create false positives because normal pattern variation may look abnormal.
- Crack-like lines should be treated as higher severity than color-only marks.
- Use local texture comparison: true defects usually break the regular local pattern.

Tile - Recommended operator action rules

Condition found	QA decision	Operator action
Consistent texture, no abnormal marks	Accept	Allow item to pass.
Removable stain or glue without damage	Rework / clean	Clean and reinspect.
Crack, fracture, permanent surface	Reject	Remove from acceptable stock.

damage		
Possible normal texture variation	Escalate	Ask for human review and compare with normal references.

Tile - RAG retrieval keywords

Use these phrases to retrieve this section: tile crack, tile glue strip, tile oil stain, gray stroke, rough tile, surface texture anomaly.

Tile - Minimum evidence to include in report

- Surface region affected and approximate size.
- Whether anomaly is crack/fracture, stain/contamination, or roughness.
- Whether cleaning/rework is possible.
- Decision and next action.

RAG answer instruction: When the retrieved section is Tile, the AI assistant must explain the visible anomaly, state whether the item is acceptable or requires review, and give the recommended operator action. Do not state a guaranteed root cause unless process data confirms it.

Toothbrush: Category-Specific QA Inspection Procedure

Product category: Personal-care product inspection. Focus areas include bristle region, brush head, neck, handle, hygiene appearance, and structural condition.

Normal accepted condition. An accepted toothbrush image should show intact bristles, clean head, correct shape, no damaged bristle cluster, no broken handle/neck, and no contamination.

Toothbrush - Inspection checkpoints

- Inspect bristle area for missing, bent, clumped, cut, or damaged bristles.
- Inspect brush head for deformation, cracks, broken edge, or foreign material.
- Inspect handle and neck for breakage, bending, or surface damage.
- Check for contamination, stains, or residues on any hygiene-critical surface.
- Check whether the product outline matches normal shape.

Toothbrush - Reject or anomaly indicators

- Reject if bristles are damaged, missing, contaminated, or unsuitable for hygiene use.
- Reject if the brush head or handle is cracked, broken, or deformed.
- Reject if contamination is visible on bristles or head.
- Escalate if bristle variation may be normal but the model confidence is low.

Toothbrush - Caution notes for the AI assistant

- Toothbrush inspection has hygiene importance, so contamination is serious.
- Small bristle spacing variation may be normal; missing or damaged clusters are defects.
- Do not allow rework if contamination cannot be safely removed or verified.

Toothbrush - Recommended operator action rules

Condition found	QA decision	Operator action
Clean and intact bristles/head/handle	Accept	Allow item to pass.
Minor packaging dust away from bristles	Rework / clean	Clean under approved process and reinspect.
Damaged bristles, contaminated head, broken handle	Reject	Remove item from customer use.
Unclear bristle region	Escalate	Request clearer close-up image.

Toothbrush - RAG retrieval keywords

Use these phrases to retrieve this section: toothbrush defect, bristle defect, toothbrush contamination, damaged brush head, personal care QA.

Toothbrush - Minimum evidence to include in report

- Affected region: bristles, head, neck, or handle.
- Hygiene impact and contamination status.
- Decision level and whether item must be rejected.
- Any need for human review.

RAG answer instruction: When the retrieved section is Toothbrush, the AI assistant must explain the visible anomaly, state whether the item is acceptable or requires review, and give the recommended operator action. Do not state a guaranteed root cause unless process data confirms it.

Zipper: Category-Specific QA Inspection Procedure

Product category: Flexible fastening component inspection. Focus areas include zipper teeth, fabric border, fabric interior, slider path, tooth alignment, and squeezed or split regions.

Normal accepted condition. An accepted zipper image should show continuous teeth, correct alignment, no missing or broken tooth, no split teeth, no damaged fabric border, and no deformation that blocks fastening.

Zipper - Inspection checkpoints

- Inspect zipper teeth for broken, missing, squeezed, split, rough, or misaligned teeth.
- Inspect fabric border and fabric interior for tears, holes, fraying, stains, or foreign material.
- Check whether the tooth line is continuous and regular along the fastening path.
- Check whether the zipper could close without obstruction based on visible geometry.
- Look for combined anomalies affecting both teeth and fabric.

Zipper - Reject or anomaly indicators

- Reject if teeth are broken, split, missing, squeezed, or misaligned enough to affect fastening.
- Reject if fabric border damage may reduce stitching strength or zipper attachment.
- Rework only if surface dirt is removable and no mechanical damage exists.
- Escalate if texture/fabric pattern causes uncertain false positives.

Zipper - Caution notes for the AI assistant

- Zipper images may include regular repeating patterns; do not treat every pattern change as a defect.
- Functional tooth defects are more serious than cosmetic fabric marks.
- Combined tooth and fabric anomalies should be treated as reject-level until reviewed.

Zipper - Recommended operator action rules

Condition found	QA decision	Operator action
Continuous aligned teeth and clean fabric	Accept	Allow item to pass.
Minor removable dirt on fabric	Rework / clean	Clean and reinspect.
Broken/split/missing/squeezed teeth or torn fabric	Reject	Remove from production use.
Ambiguous texture variation	Escalate	Send to human inspector.

Zipper - RAG retrieval keywords

Use these phrases to retrieve this section: zipper defect, broken teeth, split teeth, squeezed teeth, fabric border defect, rough zipper, zipper fastening failure.

Zipper - Minimum evidence to include in report

- Affected region: teeth, slider path, fabric border, fabric interior.

- Whether fastening function may be affected.
- Specific anomaly description and severity.
- Decision and action.

RAG answer instruction: When the retrieved section is Zipper, the AI assistant must explain the visible anomaly, state whether the item is acceptable or requires review, and give the recommended operator action. Do not state a guaranteed root cause unless process data confirms it.

Metal Nut: Category-Specific QA Inspection Procedure

Product category: Machined metal fastener inspection. Focus areas include outer surface, inner hole, thread region, corners, shape symmetry, color/oxidation, scratches, and bending/deformation.

Normal accepted condition. An accepted metal nut image should show correct shape, complete edges, clean internal thread/hole, no major scratch, no deformation, no abnormal color/oxidation, and no foreign material blocking assembly.

Metal Nut - Inspection checkpoints

- Inspect outer edges and corners for bent shape, chipped material, or deformation.
- Inspect internal hole and thread area for blockage, rough damage, missing material, or contamination.
- Inspect top and side surfaces for scratches, dents, pits, discoloration, corrosion-like marks, or oil residue.
- Check whether the nut appears flipped or oriented in a way that makes required inspection surface unclear.
- Check shape symmetry and whether the item could fit a bolt correctly.

Metal Nut - Reject or anomaly indicators

- Reject if the nut is bent, deformed, cracked, or cannot fit properly.
- Reject if internal thread/hole damage may affect fastening.
- Reject if corrosion, abnormal color, or contamination affects function or material quality.
- Escalate if a surface scratch is shallow and the model confidence is low.

Metal Nut - Caution notes for the AI assistant

- Metal reflections can cause bright regions that look like color defects; unclear reflections require review.
- Small cosmetic scratches may be lower severity unless located on functional thread or sealing surface.
- Functional geometry and internal hole integrity are more important than minor exterior appearance variation.

Metal Nut - Recommended operator action rules

Condition found	QA decision	Operator action
Correct shape, clean hole, no visible defect	Accept	Allow item to pass.
Minor removable oil/dust	Rework / clean	Clean and reinspect.
Bent shape, thread damage, corrosion-like mark, deep scratch	Reject	Remove from accepted stock and inspect tooling/process.
Reflection/lighting uncertainty	Escalate	Request image under better lighting or human review.

Metal Nut - RAG retrieval keywords

Use these phrases to retrieve this section: metal nut defect, metal nut scratch, bent nut, color defect, flip defect, thread defect, machined fastener QA.

Metal Nut - Minimum evidence to include in report

- Affected region: outer edge, corner, surface, inner hole, thread, or orientation.
- Whether fitting/fastening function may be affected.
- Presence of scratch, color change, bending, or contamination.
- Decision and recommended action.

RAG answer instruction: When the retrieved section is Metal Nut, the AI assistant must explain the visible anomaly, state whether the item is acceptable or requires review, and give the recommended operator action. Do not state a guaranteed root cause unless process data confirms it.

9. RAG Evidence and Answering Rules

The AI assistant must use this manual as a supporting knowledge source. When a user asks what the defect means or what action to take, the RAG system should retrieve the relevant category section and the general decision rules.

1. Always mention the product category when known.
2. Always separate model output from manual evidence. Example: The model detected a cable insulation anomaly. The manual states exposed conductor or cut insulation should be rejected.
3. Do not claim that the AI has proven the real manufacturing root cause from image evidence alone.
4. Use conservative language for root causes: possible cause, likely checkpoint, recommended process check.
5. If retrieved evidence is missing or weak, the assistant must say that manual evidence was not found and request human review.
6. If model confidence is low, the assistant must recommend a clearer image or human inspection.
7. If severity is critical or safety-related, the assistant must not give an accept decision without human review.

10. Standard Output Format for AI Inspection Result

Output field	Required content
Product category	Example: cable, bottle, screw, transistor, tile, toothbrush, zipper, metal nut.
Defect/anomaly status	Normal, suspected anomaly, confirmed anomaly, or uncertain.
Detected region	Location such as rim, thread, lead, surface, tooth line, bristle area, or cable insulation.
Confidence score	Model confidence as a percentage or decimal.
QA decision	Accept, rework/clean, reject, or escalate.
Manual evidence	Short retrieved sentence from this manual explaining the decision.
Recommended action	Clear operator instruction.
Human review required	Yes or no, with reason.

Example answer: The uploaded image is classified as cable with suspected cut insulation. The model confidence is 0.91. This manual states that exposed conductor or cut insulation should be rejected because it can affect electrical safety. Recommended action: remove the item from assembly, mark it as defective, and request QA confirmation.

11. Human Review and Escalation Rules

Situation	Required response
Model confidence below 70%	Escalate to human inspector. Do not issue final accept/reject as sole decision.
Image is blurry, dark, cropped, or reflective	Request a clearer image.
Critical functional area affected	Reject or hold item pending supervisor review.
RAG evidence not found	State that manual evidence was not retrieved and request human review.
Multiple anomalies detected	Use the highest severity anomaly for first decision and mention all visible findings.
Conflict between AI result and manual evidence	Prefer manual policy and escalate to human inspector.

12. Category Quick-Lookup Table for RAG

Category	Useful retrieval terms	Main risk	Default decision rule
Bottle	crack, broken, contamination, rim defect, neck defect	Structural damage or hygiene issue	Reject if cracked/broken or contaminated in critical area
Cable	cut insulation, missing wire, cable swap, bent wire, poke insulation	Electrical safety or assembly issue	Reject if conductor exposed, wire missing, or wire order wrong
Screw	thread damage, scratch head, scratch neck, manipulated front	Mechanical fit issue	Reject if thread/head/neck function is affected
Transistor	bent lead, cut lead, damaged case, misplaced	Electrical contact or placement issue	Reject if lead/case/placement is damaged
Tile	crack, glue strip, oil, rough, gray stroke	Surface quality issue	Reject cracks; clean removable contamination
Toothbrush	bristle defect, damaged head, contamination, broken handle	Hygiene or usability issue	Reject if bristles/head are damaged or contaminated
Zipper	broken teeth, split teeth, fabric border, rough, squeezed teeth	Fastening function issue	Reject if teeth/fabric damage affects fastening
Metal Nut	scratch, bent, color, flip, thread damage	Fastening or material quality issue	Reject if shape/thread/function is affected

13. Report Writing Requirements

Every generated inspection report should be concise, evidence-based, and traceable. The report must not hide uncertainty. If the AI result is uncertain, the report should clearly state that human review is required.

- Inspection ID and timestamp.
- Uploaded image filename and product category.
- Predicted status: normal, anomaly, or uncertain.
- Detected anomaly description and location.
- Confidence score and model name/version if available.
- Retrieved manual evidence summary.
- QA decision: accept, rework/clean, reject, or escalate.
- Recommended operator action.
- Human review flag and reason.
- Optional: annotated image or anomaly heatmap reference.

14. Limitations

- This manual is synthetic and intended for project demonstration and RAG testing.
- The AI assistant cannot confirm hidden internal damage that is not visible in the image.
- The AI assistant cannot guarantee root cause without process data such as machine settings, batch history, operator notes, or sensor readings.
- Dataset images may not represent every real-world lighting, background, angle, or production condition.
- Final production decisions should follow approved company QA policies and human expert judgement.

15. RAG Chunking Guidance for Developers

For best retrieval quality, split this manual by heading or category section. Each product section is intentionally written with repeated category names and defect phrases so the vector database can retrieve the correct passage from user questions.

RAG setting	Recommendation
Chunk size	500 to 1000 tokens is suitable for category sections.
Overlap	Use 80 to 150 token overlap to preserve decision rules across

	chunks.
Metadata fields	document_type, manual_id, version, category, section_title, effective_date.
Retrieval filter	When user selects category, filter by category metadata before semantic search.
Top-k retrieval	Start with k=3 to k=5. Retrieve category section plus general QA decision rules.
Answer grounding	The final LLM response must cite or quote the retrieved section summary in simple language.

16. Sample RAG Test Questions

User question	Expected retrieval and answer direction
Bottle image has a crack near the rim. What should we do?	Retrieve Bottle section and General QA Decision Levels. Expected answer: reject or escalate, because rim/neck crack affects sealing and structural quality.
Cable has a cut in the insulation. Is this serious?	Retrieve Cable section. Expected answer: reject, because exposed conductor or cut insulation is safety/function related.
The screw thread looks damaged. Can it be used?	Retrieve Screw section. Expected answer: reject if thread fit is affected; human review if unclear.
Transistor lead is bent. What is the next action?	Retrieve Transistor section. Expected answer: reject or electrical review depending severity.
Tile has an oil mark but no crack. What should happen?	Retrieve Tile section. Expected answer: clean/rework if removable; reject if permanent contamination or damage.
Toothbrush bristles look damaged. Is it acceptable?	Retrieve Toothbrush section. Expected answer: reject because bristle damage affects hygiene/usability.
Zipper teeth are broken. What does the SOP say?	Retrieve Zipper section. Expected answer: reject because fastening function may fail.
Metal nut has a color mark and scratch. What should be reported?	Retrieve Metal Nut section. Expected answer: inspect functional impact; reject if corrosion/deep scratch/thread issue; otherwise review.

17. Final Approval Note

This manual is approved for use as a prototype RAG source in the AI Defect Detection and Root-Cause Assistant project. It should be replaced or supplemented by real company SOPs, certified QA manuals, and expert-approved defect taxonomy documents before any real production deployment.

Manual status: Ready for RAG ingestion as QA_Inspection_Manual_MVTec_AD.pdf. Recommended storage path: rag_documents/QA_Inspection_Manual_MVTec_AD.pdf